TECHNICAL SHEET

Oktavia
Multi-vendor network automation

Oktavia discovers the network, checks it against the customer standard, proposes the change against declared intent and, once a person approves, applies it and confirms the finding closed. One cycle, four vendors in production, every action with a downloadable artifact.

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CONTENTS

1. Executive summary
2. The problem it solves
3. Value proposition
4. Capability categories
5. Platform modules
6. The closed loop
7. New service rollout
8. Site and device lifecycle
9. Argos, the network ChatOps
10. Security and service inventory
11. Per-vendor coverage
12. Integrations
13. Architecture and operating model
14. Platform security
15. Integration requirements
16. Implementation process
17. Licensing
18. Use cases

1. Executive summary

Oktavia is a network automation platform for enterprise IT operations. It pairs an automation engine with a unified web interface to run audits, verify configuration compliance, visualize network state, and manage changes in a controlled and traceable way.

It is built for teams that need to move fast without giving up control. Every action produces versioned artifacts, every change requires prior validation, and everything lands in a searchable history.

> **What sets Oktavia apart:** true multi-vendor in production across Cisco IOS and IOS-XE, Fortinet FortiGate and FortiSwitch, and Juniper Junos; Argos as a ChatOps layer across the whole platform; network security with configuration evidence; service inventory; and the full detect, remediate, and verify cycle on all four vendors.

2. The problem it solves

Enterprise networks grow more complex faster than the teams that run them. The symptoms repeat across almost every customer:

- **Manual audits** that take days and depend on nobody misreading a configuration.
- **Configuration drift** that goes unnoticed until there is an incident.
- **Fragmented visibility** across inventory, compliance, topology, and metrics, each in its own tool.
- **No traceability** on who changed what, when, and with whose approval.
- **Reactive operations**: the team acts when a user reports, not when the network drifts.
- **Diagnosis without closure**: audit tools find the problem, but fixing it stays manual work outside the tool.

3. Value proposition

BENEFIT	DESCRIPTION
The full cycle, not just diagnosis	Detect, propose, approve, apply, and verify the finding closed. Finding a problem is worthless if fixing it stays manual.
Audits in minutes, not days	Full compliance across the whole network, a site, a role, or a single device in one operation.
Control without risk	No change is ever applied without generated artifacts and prior validation. Each vendor is touched with its safest primitive.
End-to-end visibility	Inventory, topology, compliance, security, addressing, and configurations in a single console.
Traceable automation	Complete history of jobs, results, and downloadable artifacts in JSON, CSV, and CFG.
Genuinely multi-vendor	Each vendor has its own normalized configuration model, not a translation into Cisco syntax.
Operating by conversation	Argos audits, diagnoses, proposes configuration, and closes the loop from the chat, without switching screens.
AI with its cost in plain sight	Every answer states whether it consumed the language model or resolved deterministically, before it runs.

4. Capability categories

Oktavia capabilities group into seven categories. It is the way to read the platform by problem solved, not by screen.

#	CATEGORY	WHAT IT SOLVES
1	**Network state and configuration**	Knowing at all times what exists, how it is connected, and how it is configured.
2	**Compliance and configuration control**	Defining the standard and verifying the network meets it, with no blind changes.
3	**Remediation and service rollout**	Closing the finding and rolling out new services with approval, verification, and rollback.
4	**Network security**	Firewall and switch posture with the configuration evidence backing each finding.
5	**Observability and event automation**	Seeing operational health in real time and reacting automatically to network conditions.
6	**Intelligence and analysis**	Turning raw data into decisions: risk, drift, capacity, paths, and flows.
7	**Governance and FinOps**	Proving the return on automation and operating multi-tenant with traceability.

5. Platform modules

MODULE	FUNCTIONAL SCOPE	LICENSE
Automation center	Executive view of the network: global compliance score, devices reachable by ping and SSH, distribution by role, sites covered, active failures, and recent job activity.	Pro
Network inventory	Dynamic multi-vendor inventory with filters by site, role, platform, vendor, and site type. Visual distribution, per-device detail, audit freshness, CSV import and export with non-destructive merge.	Pro
Topology and path tracing	Interactive map with physical, control-plane, routing, OSPF, BGP autonomous system, and STP views. Per-node and per-site panels. Live path trace between two points with hop, next hop, egress interface and protocol, reverse path, and asymmetry detection.	Pro
Configuration compliance	Global and per-site score, top failures, and trends. Per-device results with severity, diagnosis, and remediation plan. Editable service catalog and rules with selective enablement by role, vendor, platform, or site type.	Pro
Desired configuration	Configuration generated from service variables and templates, with precedence by defaults, site type, site, platform, model, role, and device. JSON and CSV export.	Pro
Closed-loop remediation	Audit, propose, review the diff, approve, apply, and audit again to confirm closure. Every step leaves a downloadable artifact and lands in the history.	Pro

MODULE	FUNCTIONAL SCOPE	LICENSE
New service rollout	The same cycle for rolling out services: tunnels between sites, advertising a network in OSPF and BGP, DHCP per VLAN, routing adjustments. Each with its rollback.	Pro
Argos — network ChatOps	Plain-language assistant embedded in every screen. It understands, investigates, configures, and closes the loop. Each action states whether it consumes the AI model before it runs.	Pro
Network security	Posture computed on collected configuration across four domains: perimeter, management plane, identities, and segmentation. Global and per-device score, filters by domain and severity, evidence per finding.	Pro
Service inventory	Cross-check between what is observed on the device, what is declared as intent, and the service catalog. Classifies each service as covered, blind spot, not applied, or not applicable, with its configuration scope.	Pro
Digital Twin	Network state snapshots and comparison between desired and live configuration. Per-device drift detection, impact quantification, and an approval flow through successive states up to application.	Pro
Firewall policy analysis	Which policy decides a flow at each hop along the path, with its address translation and its logging. Resolves object groups and application sets recursively.	Pro
IP Intelligence (IPAM)	Addressing map, detected subnets, overlaps, routing neighbors, and the location of an IP address down to the access port by crossing the ARP table with the MAC table.	Pro
Configuration backups	View, compare, and download configuration files by device and date.	Pro
Advanced reports	Network health, configuration risk, drift, capacity, ports, and topology, in downloadable reports ready for audit.	Pro
Provisioning and site management	Logical structure of site types, sites, and roles defined before the first discovery run. Executive per-site view with devices, management addresses, compliance, and network configuration.	Pro
Operations and scheduled audits	Running and scheduling audits, discovery, inventory sync, and connectivity validation, with job history and artifacts.	Pro
FinOps	Automation return-on-investment calculator and operational savings analysis.	Pro
Multi-tenant management	Multi-organization, user management, and per-tenant data separation.	Pro
Real-time telemetry	Operational CPU, memory, and interface metrics stored as time series, with temporal visualization, trends, and threshold alerts.	Telemetry and Events

MODULE	FUNCTIONAL SCOPE	LICENSE
Event automation	Condition-to-action rules that react automatically to BGP session drops, OSPF neighbor drops, interfaces down, and CPU, memory, temperature, or error thresholds.	Telemetry and Events

6. The closed loop

This is what separates Oktavia from an audit tool. Finding a problem is worthless if fixing it stays manual, and applying a change blind is worthless if nobody verifies it landed right. The loop is a single path walked without leaving the platform:

1. **Audit** the chosen scope: a device, a role, a site, or the whole network.
2. **Read the compliance result**: what failed, at what severity, and which configuration lines are missing.
3. **Propose the change** against declared intent, not against a generic template.
4. **Review the diff** between desired and live configuration in the Digital Twin, with estimated impact.
5. **Approve.** A person, always. No change applies on its own.
6. **Apply** using each vendor safe primitive.
7. **Verify**: audit again and confirm the finding closed.

Every step leaves a downloadable artifact and lands in the job history. If the result is not convincing, there is a way back.

How it applies on each vendor

VENDOR	APPLY PRIMITIVE	SAFETY NET
Juniper Junos	Deferred commit confirmation over NETCONF	The device rolls back on its own if nobody confirms within the window. A session that drops mid-apply does not leave the device unreachable.
Fortinet FortiGate	Operations over the configuration API	Changes touching administrators and zones stay in permanent simulation, because moving a zone invalidates the policies referencing it.
Fortinet FortiSwitch · Cisco IOS	Secure session with vault credentials	Diff before and verification after. The rollback is the inverse change, also reviewed and approved.

7. New service rollout

The same cycle rolls out services, not just corrects deviations. You describe what you want and Oktavia proposes it as declared intent:

- **A tunnel between two sites**, with the routes that run through it, and its complete teardown when the project ends.
- **Advertising a network**: logical interface, OSPF announcement, access control lists, and BGP on the site edge devices, with ping verification.
- **Editing routing**: router identifier from the logical interface, recommended site summarization.
- **DHCP for a VLAN**, with the range proposed from what Oktavia already knows about the segment.
- **Adopting what exists**: read a device backup and declare as intent what it already has configured, so auditing can start without redesigning anything.

> Every service is proposed, edited before applying, and ships with its rollback. Nothing is written to the network without a person approving it.

8. Site and device lifecycle

Opening and closing sites, and onboarding, replacing and retiring devices, are one-button flows with preview, approval and verification. Each is first rehearsed on the test network (Cisco Modeling Labs), where Oktavia creates the nodes with a minimal day 0 and wires them; powering them on is always the operator's call.

- **Site creation**: the design of its site type is cloned, with proposed and editable addressing, the hub side (BGP, tunnel, subinterface, VLAN), and the lab nodes and test PCs.
- **Site removal**: cleans the hub in order (BGP `no neighbor`, tunnels, subinterfaces, VLANs and trunks), cleans Oktavia and the lab, and validates that nothing of the site is left. Create, remove and create again is repeatable.
- **Device onboarding**: access with the vault credential, discovery and audit, configuration from intent, telemetry, loop and verification.
- **Device replacement**: a different model under the same name, editable port mapping, link cutover, and the old device retired once the new one is powered on and in service (the practice is to verify it before retiring the old one). The new SSH key is accepted only with operator confirmation.
- **Device retirement**: a downloadable archive package without secrets, and cleanup of intent, inventory, compliance and telemetry.

On Cisco SD-WAN the same loop goes through the Manager: edge onboarding with its Config Group and bootstrap, and creation and removal of LAN segments. The details are in the article Test before you touch production: the Oktavia test network.

9. Argos, the network ChatOps

Argos is the assistant embedded in every Oktavia screen. You ask in plain language and it answers with network data, not generalities. It covers four moments of daily work:

- **Understand**: network state, compliance, what to address first, diagnosis of a device or a site.
- **Investigate**: path between two IP addresses forward and back, correlation of failures against configuration, anomaly and unreachable device detection.
- **Configure**: describe the service in plain language or paste the configuration, and Argos proposes it as declared intent for the operator to edit before applying.
- **Close the loop**: audit, read compliance, take it to the Digital Twin, approve, and verify. Each step is a button, not a sentence you have to type.

Two design principles separate it from a generic assistant. Every answer states whether it consumed the language model or resolved deterministically, so the cost is visible before you click. And no action that touches the network happens without a person approving it.

What uses AI and what does not

Most of what Argos does never touches the model: they are queries and calculations over already collected data, with reproducible results. The model steps in when something has to be written, prioritized, or interpreted from free text.

CAPABILITY	ENGINE
Network state, compliance, diagnosis by device or site	Deterministic
Service inventory and blind spots	Deterministic
Security posture and its findings with evidence	Deterministic
Path between two IP addresses, forward and back, with firewall policy per hop	Deterministic
Locating an IP address down to the access port	Deterministic
Importing a device configuration as declared intent	Exact parsers where they exist; AI for the rest
Closing the loop: audit, compliance, Digital Twin, verify	Deterministic
Starting from a ticket: recognizing the sites in the text, auditing their configuration and state, and summarizing the findings	Deterministic
Interpreting an already computed path: architecture and troubleshooting findings	AI
Explaining security: real priority and concrete per-vendor changes	AI
Recommending what to declare first from the service inventory	AI
BGP and OSPF routing analysis: state, risks, and recommendation	AI
Configuring by conversation: describing a service in plain language	AI
Correlating compliance failures with the configuration	AI only when there are findings

CAPABILITY	ENGINE
Optimizing scopes and proposing a standard	AI on top of a deterministic calculation

The pattern repeats: the data is computed first, and the model only writes or prioritizes on that result. It never invents findings, it works with the ones the calculation produced.

10. Security and service inventory

Network security

Security posture computed on the collected configuration of firewalls and switches, across four domains: perimeter, management plane, identities, and segmentation. Each finding carries the configuration line backing it and its recommendation in the vendor own syntax. It includes a global and per-device score, and filters by domain and severity.

Vulnerabilities by version

Each device's software version is matched against NVD, CISA KEV and FIRST EPSS: actively exploited vulnerabilities first, with a rating per device, site and tenant, the minimum safe version per model that feeds the software standard, and a printable CVE report per network or site. Lab images are marked as approximate and do not count as real exposure. More in the article Security in Oktavia and Argos.

Service inventory

It answers the question that comes before compliance: what is configured on each device, and how much of it is declared as intent. It cross-checks what is observed against what is declared and against the service catalog, and classifies each service into four states, along with the configuration scope that defines it.

STATE	MEANING
Covered	The service is configured on the device and declared as intent. Oktavia audits it.
Blind spot	It is configured on the device but nobody declared it. Nobody audits it, and a change to it goes unnoticed.
Not applied	It is declared as intent but does not appear in the device configuration.
Not applicable	It does not correspond to the platform or the role of that device.

From the inventory itself, blind spots are declared as intent in a single step, without touching the network: what was already there simply starts being audited.

11. Per-vendor coverage

VENDOR	ACCESS METHOD	COVERAGE
Cisco IOS / IOS-XE	SSH	Discovery, compliance, desired configuration, remediation
Fortinet FortiGate	REST API	Discovery, compliance, desired configuration, remediation, firewall policies
Fortinet FortiSwitch	SSH	Discovery, compliance, desired configuration, remediation
Juniper Junos	NETCONF	Discovery, compliance, desired configuration, remediation, zone policies

> **Coverage grows with every project.** The architecture is multi-platform by design: each vendor has its own normalized configuration model, not a translation into Cisco syntax. Virtual domains and interfaces in FortiOS, contexts and zones in Junos. Integrating a new platform is four known pieces: discovery adapter, normalizer, compliance rules, and generation templates. It is bounded, repeatable work, done against the customer real installed base, in proofs of concept and projects.

12. Integrations

Oktavia was designed to coexist with the customer ecosystem, not to replace it. It can feed and be fed by the tools already in operation.

INTEGRATION	STATUS	USE
SNMP and metric collection	Production	Native telemetry engine: polling and storage in a time-series database.
Grafana	Compatible	Oktavia time-series database is a direct Grafana datasource; the customer builds dashboards on the same metrics with no extra transformation.
Zabbix and other SNMP systems	Interoperable	Devices exposed over SNMP can be consumed by whichever monitoring system the customer already runs.
Email (SMTP)	Production	Notifications and alerts by email over any customer SMTP server.
Anthropic (Claude API)	Production	AI engine for analysis, recommendations, and conversational assistance.
HTTP webhooks	Production	Event delivery to any HTTP integration point, which is the way to connect ticketing and messaging tools.

INTEGRATION	STATUS	USE
External source of truth	Optional	Inventory ingestion from an external system of record if the customer already runs one.
CSV and REST API	Production	Import and export of inventory, catalogs, and data across every module; programmatic integration over API.

Integration differentiator: Oktavia contributes the intent and compliance layer, the standard the network is compared against, which a metrics dashboard or a traditional monitoring system does not have. They measure; Oktavia also validates against a standard and automates the response.

13. Architecture and operating model

The platform separates the web interface from the automation engine. The interface owns the user experience, tenant organization, and persistence; the engine executes against the devices. They talk over a REST API authenticated with a key.

- **Persistence:** relational database with per-tenant data separation.
- **Artifacts:** JSON, CSV, and configuration files, downloadable and versioned.
- **Deployment:** orchestrated containers, on customer premises or in a private cloud.
- **Device access:** SSH, NETCONF, and native vendor API, with credentials encrypted in a vault.

Operating model

- Everything runs as **asynchronous jobs** with complete history.
- Every job produces **versioned artifacts**, downloadable.
- **No direct changes are ever applied** without generating and validating artifacts first.
- Remediation actions are **supervised and traceable**, with an identified approver.

14. Platform security

Oktavia has read access and, when the customer authorizes it, change access to their network. Its own security is part of the scope, not an annex.

Where the data lives

The platform is deployed on the customer infrastructure, on premises or in their private cloud. Inventory, configurations, and artifacts do not leave it. The only exception is AI analysis: when an action that consumes the model runs, the content that query needs travels to the AI provider. Argos states before every action whether it will consume it, so the decision is the operator's, case by case, and the deterministic capabilities cover most of the work without sending anything.

Device credentials

They are stored encrypted at rest in the database, with a key that lives in the deployment configuration and not in the code. They are never shown in the interface nor written to logs. We recommend read-only

credentials for discovery, and separate change credentials only if remediation is in scope.

Access to the platform

Mandatory login, integrated with the corporate identity provider. On top of that, an access allowlist by IP address or network, managed from the interface itself, with two modes: **monitor**, which logs rejections without blocking, and **strict**, which blocks. Switching to strict is guarded: it cannot be enabled if the operator own address is not covered. The same list applies to the web interface and to the automation engine.

Data separation

Multi-tenant with row-level separation: every query is scoped to the user organization. One customer data is not reachable from another customer session, not even through a programming mistake.

Between components

The web interface and the automation engine authenticate with a key on every call, in both directions. The engine exposes no operation without it.

No change without approval

This is a security property too, not only a process one: no action that writes to the network happens without a person approving it, and every application is recorded with its approver, its diff, and its artifact.

Corporate identity and second factor

Login integrates with the customer identity provider over OpenID Connect. Oktavia supports **Okta**, and with it single sign-on and whatever second factor the provider policy already enforces: onboarding, offboarding, and passwords live where the corporate directory already lives, with no parallel user list to maintain. Local passwords are kept for contingency only.

The platform can also require the authentication to be recent, not merely that a session exists at the provider: it can demand a fresh authentication event, restrict the accepted methods, a passkey for example, and require a specific assurance level, such as the Okta two-factor one. That is what makes closing the browser ask for the second factor again.

> **Confidentiality.** Handling of customer information is governed by the confidentiality agreement signed at the start of the project, with the scope, retention, and return or destruction of the data agreed there.

15. Integration requirements

To deploy Oktavia over an existing network we ask the customer for:

- Base device inventory.
- List of sites and site types.
- Operational roles per device.
- Vendor, platform, and version, where the data exists.

- Management addresses and access method.
- Read credentials for discovery.
- Change credentials, if remediation is in scope.
- Backups or available live configuration.
- Current configuration standards.
- Operational constraints and change windows.
- Approvers and technical contacts.

Minimum data per device

Identity (hostname, management address, site, role), operating system and model, interfaces, layer 2 discovery neighbors, BGP and OSPF neighbors where they exist, live configuration or backup, and the baseline operational services: NTP, DNS, SNMP, SSH, banners, and time zone.

Preferred order for sourcing the data

1. Native vendor API.
2. NETCONF, RESTCONF, or remote procedure calls.
3. Structured vendor parsers.
4. Command-line output parsing.
5. Live configuration or backups.

Server for Oktavia

Oktavia runs on a single Linux x86_64 server (Ubuntu 22.04 or 24.04 LTS recommended) with Docker and Docker Compose v2, physical or virtual, on premises or in the customer's cloud. The figures are estimates for sizing the machine: they depend on the number of interfaces per device, how often audits run, and how long telemetry is kept.

Oktavia Pro includes inventory and discovery, configuration and state compliance, the Digital Twin with approved remediation, backups, and Argos.

DEVICES	VCPU	RAM	SSD DISK
Up to 20 (proof of concept)	4	8 GB	60 GB
20 to 100	4	16 GB	100 GB
100 to 300	8	16 GB	150 GB
300 to 1,000	16	32 GB	250 GB
More than 1,000	Sized with the customer (separate audit workers and database)		

Oktavia Pro with Telemetry and Event Orchestration adds SNMP collection every 30 seconds, syslog reception, the time-series database with 7 days of retention, and the always-on event evaluator.

DEVICES	VCPU	RAM	SSD DISK
Up to 20 (proof of concept)	4	8 GB	80 GB
20 to 100	8	16 GB	150 GB
100 to 300	8	32 GB	250 GB
300 to 1,000	16	64 GB	500 GB
More than 1,000	Sized with the customer (separate time-series database)		

> **Proof of concept with 10 or 20 devices:** a single virtual machine with 4 vCPU, 8 GB of RAM, and an 80 GB SSD covers Oktavia Pro and telemetry. With 10 devices there is plenty of headroom. In our lab the full platform uses about 1.5 GB of RAM at rest; the rest is headroom for parallel audits, the operating system, and growth.

Considerations

- Telemetry disk grows with the interfaces per device and with retention: keeping 30 days instead of 7 multiplies that part by four.
- Backups and audit results grow with audit frequency; the retention policy is agreed during implementation.
- Argos with a cloud AI provider needs no extra resources. A local model needs at least 16 GB of additional RAM and, preferably, a GPU.
- Connectivity: the server must reach the devices' management plane (SSH, HTTPS/API, NETCONF, and SNMP depending on the vendor) and receive syslog (UDP 514) if telemetry is enabled. Users reach the console over HTTPS.

16. Implementation process

1. **Data onboarding.** Receive the base inventory, normalize names, roles, sites, platforms, and models, and load the initial file.
2. **Controlled discovery.** Validate reachability, run discovery in batches, collect backups, and normalize the observed inventory.
3. **Intent baseline.** Define services per scope, load service variables, enable compliance checks, and configure the customer service catalog.
4. **Validation.** Run the full audit, review the dashboards, and tune rules and exceptions.
5. **Operation and remediation.** Propose the change, simulate it, approve it, apply it, and recompute compliance to confirm closure.

Estimated time to a working deployment

NETWORK SIZE	ESTIMATED TIME
Small network	1 to 3 days
Medium network	3 to 7 days
Large or complex multi-vendor network	2 to 4 weeks

> AI accelerates the preparation: inventory cleanup, role and platform normalization, parser prototyping, and catalog generation. Final validation of intent, compliance, and changes stays operational and human.

17. Licensing

The complete platform is **Oktavia Pro**. Continuous telemetry and event orchestration are licensed separately because they have a different consumption profile, they store time series and run a permanent evaluator, and not every network needs them on day one.

Oktavia Pro

Multi-vendor discovery, inventory, configuration and state compliance, desired configuration per device and per scope, the full remediation cycle with Digital Twin, new service rollout with rollback, topology with path tracing and firewall policy analysis, network security, service inventory, IP Intelligence, configuration backups, advanced reports, provisioning, scheduled operations, FinOps, multi-tenant, and the complete Argos.

Telemetry and Event Orchestration

Real-time metrics, telemetry dashboards compatible with Grafana, and the event automation engine: condition-to-action rules that react automatically to BGP session drops, OSPF neighbor drops, interfaces down, and CPU, memory, temperature, and error thresholds.

Included with every license: support and updates

While the license is active, every Oktavia deployment receives the platform's security patches, updates and new features, at no extra cost.

18. Use cases

- Configuration compliance audit across multiple sites in minutes.
- Automatic detection of configuration drift after a maintenance window.
- Desired configuration generation for new devices by role and by site.
- Live path trace between two points with next hop, egress interface, and asymmetry detection.
- Capacity visibility on wide area interfaces before saturation.

- Post-maintenance connectivity validation across the whole network.
- Change control with a diff between desired and live configuration before applying.
- Onboarding a new network with site types and sites configured before the first discovery run.
- Firewall security posture with the configuration evidence backing each finding.
- Discovering which services are configured and audited by nobody, and declaring them without touching the network.
- Knowing whether a flow passes: which firewall policy decides it at each hop, with its address translation and its logging.
- Locating an IP address down to the access port by crossing the ARP table with the MAC table.
- Adopting the configuration a device already has as declared intent, to start auditing it without redesigning anything.
- Fixing a deviation end to end: propose, approve, apply, and verify it closed.
- Bringing up a tunnel between two sites with its routes, and tearing it down completely when the project ends.
- Advertising a new network in OSPF and BGP on the site edge devices, with ping verification included.

About Oktanet

Oktanet is a Mexican network automation company focused on controlled, traceable, and scalable operations for enterprise environments in Latin America.

For a demonstration on your own environment, reach us at oktanet.io.

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